AN ECLECTIC TECHNOLOGIES DETERMINATION UNDER FD-BL

FD-BL-D1 — Mode of Discharge: Pre-Authorization by Verified Gatekeeper

Determination under FD-BL

Version 0.5 — DRAFT · 2 October 2026

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Designation: FD-BL-D1
Class: Determination under FD-BL (not a Foundational Definition — see §2)
Governing instrument: FD-BL — Foundational Definition — The Bright Line. This instrument determines the question FD-BL carried at §6.1, which FD-BL now closes to a pointer here. Versions are held in FD-REG.
Version: 0.5 — DRAFT (issued in the October 2026 Edition; supersedes v0.4)
Date: 2 October 2026
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Organization: Eclectic Technologies
Companion constructions: CB Proposer–Gate Pattern; DBA-VC Vendor Companion (consequence classifier); DBA-GAS-FC2 (whose construct is named there "operating-envelope-with-setback"; the name is quoted, and in this family's terms the region is a pre-authorized envelope under FD-EV)

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1. Purpose

FD-BL §4.5 requires that an action placed above the Bright Line have substantive, attributable human authorization by a qualified Command Broker, enforced architecturally, and that the requirement is not discharged by a procedure instructing an operator to confirm. FD-BL §6.1, as it stood at v0.2, left one question open: whether that authorization may, for a narrow and specified class of actions, be discharged **in advance** rather than in the moment. This instrument determines that question.

This determination resolved FD-BL §6.1. Since FD-BL v0.3, §6.1 has been a pointer to this instrument, which carries the resolution and its amendment set.

2. Authority, class, and what this does not alter

2.1 Class

FD-BL-D1 is a **determination**, not a foundational definition. By the family test at FD-BL §2.2, a concept belongs to the Foundational Definitions family only when the correctness of other documents depends on agreeing what *kind of thing* it is. This instrument does not define a kind of thing; it resolves the **mode** by which an existing obligation may be discharged. It is therefore designated as a determination bound to FD-BL rather than as a peer of FD-BL or FD-LD.

2.2 What this determination may and may not do

This determination operates under the constraint FD-BL §6.1 placed on it at v0.2, which FD-BL §6 records as honored and which is restated here as binding:

It varies the mode of discharge of the FD-BL §4.5 authorization. It does not vary placement (FD-BL §4.2), granularity (FD-BL §4.3), or scope (FD-BL §4.4). An action eligible for advance discharge under this instrument remains an above-line action, placed on the consequence of the individual action, produced by a component within scope of the obligation. Nothing here reclassifies any action to below the line, and nothing here creates an intermediate class of action.

2.3 The governing principle it applies

Discharge is a verification question, governed by FD-BL §4.5: the Command Broker is compensation for unverifiability, and where an authorization decision is itself **bounded by proof** the compensation is not required. This is the same principle by which a deterministic protective function performs the highest-consequence actions without a broker (FD-BL §4.5): what it will do is known before it does it, so judgment is not required where proof is available. This determination extends that principle from a native protective function to a *constructed* verified element, under conditions that make the construction's decision equally bounded by proof.

3. The criterion: bounded by proof, in both parts

An authorization decision about a proposed action decomposes into two parts, and the criterion applies to **both**:

- **Admissibility** — is the action within a pre-authorized, verified region?
- **Selection** — which action, at what magnitude, in what order, at what rate, within that region?

A construction that bounds admissibility but leaves selection free **has not** produced an authorization decision bounded by proof, because the selection remains unbounded (as FD-BL §6.1 recorded at v0.2). Advance discharge is available only where a construction bounds **admissibility and selection together**. The conditions in §4 are the conditions under which that holds.

4. The determination

The eligible class is defined here. It is the class of above-line actions for which the authorization decision is expressible as a deterministic check on measured state, evaluated by a formally verified

element that the proposing component cannot modify and cannot bypass. FD-BL §6.1 described this class at v0.2; this section now carries the definition and FD-BL §6.1 points to it.

Advance discharge of the FD-BL §4.5 authorization is permitted for an action in the eligible class if and only if the construction satisfies all of R1, R2, and R3.

4.1 R1 — The whole safety case lives in a verified gatekeeper

The admitting element is a deterministic, formally verified **gatekeeper** through which every above-line action in the class must pass, and which the **proposer**, an in-scope component under FD-LD §4.3, can neither modify nor bypass. The proposer holds **proposal authority only**; the gatekeeper holds **veto authority** and is the sole path by which any proposal reaches the process. Because the entire safety case lives in the gatekeeper, the proposer's unverified behavior, whether non-deterministic or merely not formally tested, is irrelevant to safety: it may be opaque, probabilistic, or occasionally wrong, and still cannot cause an unsafe action, because every action it proposes is disposed of by a component that carries the safety case.

4.2 R2 — Completeness over state, trajectory, and rate (this is what bounds selection)

The gatekeeper's acceptance set shall be **proven** to be a subset of the truly-safe set over **state, trajectory, and timing/rate** — not a setpoint range check. Soundness requires that every admitted action is genuinely safe in the current state, on the current trajectory, at the current rate. Completeness requires that this holds across the whole domain the proposer can explore, and that **no admissible sequence, ordering, or rate of admissible actions can compose into an unsafe path** (anti-ratcheting and anti-oscillation). Only where this proof exists is **selection**, and not merely admissibility, bounded by proof. **Where the proof cannot be produced, advance discharge is not available and the action takes in-the-moment Command Broker authorization under FD-BL §4.5.** An acceptance set shown only over static bounds is a range check and does not satisfy R2.

What one action is. The actions R2 composes are individuated under FD-BL §4.3, which defines an action and provides that where the parts of a change would place lower than the whole, the whole is placed. FD-BL §4.3 decides what one action is; R2 governs how actions so individuated compose.

4.3 R3 — Envelope changes are categorically excluded from advance discharge

An action that changes the pre-authorized envelope itself — or the gatekeeper's configuration, limits, or logic, or that moves the pre-authorized envelope toward a protective setpoint — is **not** eligible for advance discharge under any construction. No runtime gatekeeper can validate a change to its own safety basis. Such an action is an above-line action that requires either in-the-moment Command Broker authorization or an offline requalification that produces a new verified gatekeeper artifact before it is fielded. A live-learning element that updates the gatekeeper in the field is prohibited; any retrain is an offline event producing a new artifact that re-enters requalification.

4.4 What changes, and what does not

Under R1–R3 the human authorization is **not skipped; it is relocated in time** to the point at which a qualified human, the Command Broker discharging in advance (FD-BR §4.1, §4.2), (a) defined and authorized the verified envelope, including any setback from protective setpoints; (b) accepted the completeness proof required by R2; and (c) retains sole authority over any change to the envelope under R3. Every runtime action in the class is still authorized — by a proof a qualified human accepted — exactly as a protective function's actions are authorized in advance by its qualification. Placement,

granularity, and scope are untouched (§2.2). Where R2 fails or R3 applies, the obligation is discharged in the moment, unchanged from FD-BL §4.5.

5. The two-object model (settling FD-BL §6.1's explicit question at v0.2)

FD-BL §6.1, as it stood at v0.2, asked whether the consequence classifier and the verified check are one object or two. **They are two distinct objects, in series, with distinct obligations, and shall not be merged.**

- The **consequence classifier** answers **placement** (FD-BL §4.2–§4.3): for each action, is it above or below the line? It routes action by action. It is not a safety gate and does not authorize.
- The **gatekeeper** answers **mode of discharge**: for an above-line action, is it inside a valid pre-authorized envelope satisfying R1–R3, such that its FD-BL §4.5 authorization may discharge in advance? It vetoes; it does not classify.

Classification occurs first; the gatekeeper applies only to the above-line actions, and only where a valid envelope exists. Merging the two collapses a placement question into a discharge question and reintroduces the composite-classification error foreclosed at §6. A supplier shall implement them as two objects, and the evidentiary record shall show both the classifier's routing basis and the gatekeeper's acceptance-set proof.

6. Foreclosures retained

Two routes are foreclosed by FD-BL §6.1 and remain foreclosed; this determination does not reopen them:

- **Composite classification is foreclosed.** Classifying a system by the properties of the composite (in-scope component plus verified supervisor) rather than of the generating component is a change to placement — it makes architecture the measure of where the line falls, the same structural error as the technology-class and layer formulations. This determination changes mode, not placement; the action remains above the line, and the gatekeeper *admits* the action under an authorization the Command Broker gave in advance, and does not reclassify it.
- **A third tier is foreclosed.** An intermediate classification with architectural entry criteria makes architecture determinative of placement. It is foreclosed on criterion-integrity grounds, overriding the assessment that it is the cleaner route: containing a change by defining a category on the wrong property preserves the defect it was meant to avoid.

7. Sector instantiation requirements

This determination is sector-neutral. Sector annexes shall instantiate it, and shall in particular specify:

- **Denied-condition immutability.** Under denied conditions the pre-authorized envelope shall become **immutable**, and re-engagement shall require a deliberate, authenticated act rather than a proposal. This determination requires the annex to specify the denied-condition behavior; it does not itself set qualification grade. For nuclear, the protection-grade qualification bar and the behavior under denial belong to the nuclear annex.
- **Reach of the obligation preserved.** An above-line action, placed on its own consequence under FD-BL §4.2, does not fall outside the obligation because an in-scope component's contribution to it was indirect or partial. An above-line action that an in-scope component initiated or materially influenced is within the obligation's reach, whatever other components or persons also

contributed to it (CB-Annex-Nuclear). “Materially influenced” is a test of reach, not of placement: it decides whether the obligation reaches an action already placed above the line, and never whether an action is above the line. Reliance by a person on output of an in-scope component that makes no change of its own (FD-BL §4.2) is not material influence: the action the person then takes is that person’s, and the output carries the obligation of FD-BL-D2. Material influence reaches cases in which the in-scope component’s output would itself make the change, including through its acceptance (FD-BL §4.3). Advance discharge does not narrow that reach. Such an action is eligible for advance discharge only through a gatekeeper satisfying R1–R3, and otherwise takes in-the-moment authorization.

- **Qualification for the advance mode.** The person who authors, reviews, and accepts a pre-authorized envelope, and who accepts the R2 completeness proof, is the Command Broker discharging the FD-BL §4.5 obligation in advance (FD-BR §4.1, §4.2), and shall be qualified for that mode. The competence it requires is a mode-specific competence within the Command Broker qualification, distinct from the competence for in-the-moment authorization; the qualification standard shall address both within the one role.

8. Accountability

Accountability for actions discharged in advance rests with the **named Command Broker who authorized the envelope and accepted its proof, discharging in advance, and with the asset owner-operator** — the same allocation FD-BL §4.5 makes for in-the-moment authorization. A supplier may build and verify the gatekeeper; a supplier may not hold the accountability, and no software component holds it. Any change to the envelope under R3 re-invokes this accountability at the point of requalification.

9. Relationship to adjacent open items

- **Indication integrity (FD-BL §6.2 / FD-BL-D2).** Resolved. A gatekeeper is bounded by proof only if its inputs are trustworthy; a gatekeeper acting on corrupted or drifted indication is not bounded by proof. The requirement that the measured state on which the gatekeeper’s deterministic check operates be traceable to independently verifiable values — stated on an interim basis in the v0.1 issue of this determination — is now settled generally by **FD-BL-D2 (indication integrity)**, which binds both this gatekeeper-input case and the human-broker case under one rule. Advance discharge under R1–R3 is valid only on inputs satisfying FD-BL-D2, I1–I3. See FD-BL-D2.
- **Terminology reservation (FD-BL §6.3 / FD-EV).** Resolved. This determination uses “pre-authorized envelope” for the proof-bounded region within which advance discharge is valid — the authorization sense (Sense G) now reserved by **FD-EV**. The operational sense of “envelope” (a region whose violation is a runtime event to detect) is renamed “operating band” by FD-EV; this determination’s usage is conforming and unchanged. See FD-EV.

10. Conformance and amendment set

This determination governs the mode of discharge of the FD-BL §4.5 obligation. Instruments carrying operative Command Broker requirements shall conform to it and shall **propagate by reference** (they cite FD-BL-D1; they do not restate it). The amendment set, carried in the Pass 2 conformance drafts:

- **CB Proposer–Gate Pattern** — cited as the architectural basis; its Condition I (completeness over state/trajectory/rate) and Condition II (action-within vs change-to the envelope) are made normative via R2 and R3.
- **Command Broker Implementation Guide; CB-IB Qualification Standard** — add the pre-authorization mode and the advance-mode competence of the Command Broker who authors the envelope (§7).
- **CB-CT Commissioning and Joint Training** — add the gate for authoring, reviewing, and approving a pre-authorized envelope and the accountable party for one; redirect the material-change test so that an envelope change routes to requalification rather than broker re-qualification.
- **ISA Parent Framework §6.2/§6.3; IEEE V&V §5.3** — align operator-as-executor and supplier-conformance language with the two-object model (§5) and R1–R3.
- **CB-Annex-Nuclear §3.2/§3.4** — retain the materially-influenced test; specify denied-condition immutability (§7).
- **DBA-W §5.5.4** — couple to FD-BL §6.2, now determined by FD-BL-D2 (gatekeeper inputs and indication integrity).
- **62443 AI Gap Analysis FR 2** — add, to the use-control taxonomy proposed to the standards committee, a category for proof-bounded pre-authorized autonomous operation, distinct from prohibited-autonomous and operator-guided.
- **DBA-GAS-FC2; DBA-ES-GN-FC8** — harvested as instantiation and evidence (the construct DBA-GAS-FC2 names "operating-envelope-with-setback," quoted as that instrument's name; the firmware-trust precondition on the carve-out), not merely as recipients.

11. Open at this version

- Methods for demonstrating R2 completeness (state/trajectory/rate) tightly enough to be accepted rather than assumed, for high-dimensional proposers, are not standardized; this determination requires the proof but does not prescribe the method.
- The proposer–gatekeeper interface contract (what a proposal must contain; how vetoes are surfaced; how proposer liveness is reconciled with the gatekeeper's safety obligation) is left to the Command Broker specification.
- Sector qualification bars for the advance mode and for denied-condition behavior are left to the sector annexes.

Revision History

0.5 — 2 October 2026. Issued in the October 2026 Edition, and recorded in FD-REG v0.22 in the same act. One cumulative entry, extended at each step of the Action Plan for the October 2026 Edition that opens this instrument (Steps 3 and 4).

Step 3, 2 October 2026 (F-01; F-08 pointer; F-12, F-18, F-22, F-23; F-25 and F-24 checked). **F-01:** the second bullet of §7 is recast as a reach clause. An above-line action, placed on its own consequence under FD-BL §4.2, does not fall outside the obligation because an in-scope component's contribution was indirect or partial, and "materially influenced" is stated to be a test of reach and not of placement. FD-BL §4.2 is cited for placement only. DBA-ES is dropped as authority, because FD-BL §7 lists it as still restating the technology-class formulation; CB-Annex-Nuclear is kept. The clause stays in §7, where sector annexes look for it. **F-08:** R2 points to FD-BL §4.3 for what one action is, and states that FD-BL

§4.3 individuates and R2 composes. **Scope after Decision B:** R1's proposer, §4's eligible class and §6's composite are stated for an in-scope component under FD-LD §4.3 rather than a generative one. **F-12:** §4 now defines the eligible class itself, and §1, §2.2, §3 and §5 describe FD-BL §6.1 in the past frame, as it stood at v0.2. **F-18:** "operating-envelope-with-setback" in the header and §10 is annotated as DBA-GAS-FC2's quoted construct name. **F-23:** R1 to R3 are numbered §4.1 to §4.3, keeping their R labels, so that §4.4 follows in sequence. **F-22:** the date line follows the working-master convention and takes the issue date at issue. **F-25:** the acronyms MOC, APC and BES do not occur in this instrument, so none is expanded. **F-24:** the body carries no provenance narrative of the kind the review named; §1's account of how FD-BL §6.1 was reduced to a pointer is restated as fact. The first and third bullets of §7 and the whole of §8 are untouched; they belong to Step 4.

Step 4, 2 October 2026 (F-03 on FD-BL-D1's side). On the author's determination of 2 October 2026 (Decision C, Option C1), the envelope author is the Command Broker discharging in advance. §7's third bullet is retitled "Qualification for the advance mode" and makes the competence a mode-specific competence within the Command Broker qualification, addressed within the one role. §8 names the Command Broker discharging in advance as the accountable human, and §10 and §11 speak of the advance-mode competence in place of "the envelope author." §4.4 names the qualified human as the Command Broker discharging in advance. The verb is aligned: R1 calls the gatekeeper the admitting element rather than the authorizing element, and §6 says the gatekeeper admits under an authorization given in advance rather than supplying the authorization. The first bullet of §7, on denied-condition immutability, is unchanged.

Verification read, 2 October 2026. §2.2 and §5 cite "the FD-BL §4.5 authorization" in full, because this instrument has no §4.5 of its own. No change to placement, granularity, scope or obligation.

Review of 2 October 2026. On the author's instruction to apply the cold review of the October 2026 Edition review draft as recommended (FD-BL Reconciliation/Independent_Review_2026-10-02_FD_October_2026_Edition_Review_Draft_v0_1), this instrument takes findings G-05, G-06, G-08, G-09, G-10; each edit is quoted in the review's application record.

0.4 — September 2026. Governing-instrument line conformed to the citation-form determination: FD-BL is named without a version token and the version is left to FD-REG. The previous line cited FD-BL v0.2 §6.1, a section that FD-BL v0.3 had closed to a pointer, so the citation directed a reader to the superseded text because that was the only text in which the cited section still stood. No change to R1–R3, the two-object model, the foreclosures, or the amendment set.

0.3 — August 2026. Conformed the §9 terminology-reservation bullet to FD-EV: the envelope hold is discharged, the reserved "pre-authorized envelope" usage is confirmed conforming, and the operational sense's rename to "operating band" is cited. No change to R1–R3, the two-object model, or the amendment set.

0.2 — August 2026. Conformed §9 to FD-BL-D2 (indication integrity): the interim indication-integrity floor is superseded and now points to FD-BL-D2, which settles the gatekeeper-input and human-broker cases under one rule. No change to R1–R3, the two-object model, the foreclosures, or the amendment set.

0.1 — August 2026. Initial issue, resolving FD-BL v0.2 §6.1. Determines that advance discharge of the §4.5 authorization is permitted for the narrow class, under R1 (whole safety case in a verified gatekeeper), R2 (proven completeness over state, trajectory, and rate — bounding selection, not only admissibility), and R3 (envelope changes categorically excluded); states the two-object model

(consequence classifier and gatekeeper in series); retains the composite-classification and third-tier foreclosures; requires denied-condition immutability at the sector annex; and sets the amendment set for Pass 2.